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### 2.1.1. List operations

08:32

Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options:

1. Add
2. Remove
3. Display
4. Quit

The program should repeatedly prompt the user to enter a choice from the menu. Depending on the choice selected, the program should perform the following actions:

- **Add:** Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input".
- **Remove:** Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it; otherwise, display "Element not found". If the list is empty, display "List is empty".
- **Display:** Displays the current list of integers. If the list is empty, display "List is empty".
- **Quit:** Exits the program.
- The program should handle invalid menu choices by displaying "Invalid choice". Ensure that the program continues to prompt the user until they choose to quit (option 4).

```
# listOps.py
```

```
int_list = []
```

```
while True:
```

```
    print("1. Add")
```

```
print("2. Remove")
```

```
print("3. Display")
```

```
print("4. Quit")
```

```
choice = input("Enter choice: ")
```

```
if choice == '1': # Add
```

```
    value = input("Integer: ")
```

```
    try:
```

```
        num = int(value)
```

```
        int_list.append(num)
```

```
        print("List after adding:", int_list)
```

```
    except ValueError:
```

```
        print("Invalid input")
```

```
elif choice == '2': # Remove
```

```
    if not int_list:
```

```
        print("List is empty")
```

```
    else:
```

```
        value = input("Integer: ")
```

```
        try:
```

```
            num = int(value)
```

```
            if num in int_list:
```

```
                int_list.remove(num)
```

```
                print("List after removing:", int_list)
```

```
            else:
```

```
                print("Element not found")
```

```
        except ValueError:
```

```
            print("Invalid input")
```

```
elif choice == '3': # Display
```

```
    if not int_list:
```

```
        print("List is empty")
```

```
    else:
```

```
        print(int_list)
```

```
elif choice == '4': # Quit
```

```
    break
```

```
else:
```

```
    print("Invalid choice")
```

Average time  
**0.070 s**  
66.47 ms

Maximum time  
**0.100 s**  
100.00 ms

2 out of 2 shown test case(s) passed  
1 out of 1 hidden test case(s) passed

Test case 1 passed Setup Run

Expected output

Actual output

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 1

Integer: 10

List after adding: [10]

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 3

Integer: 10

List after adding: [10, 10]

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 2

Integer: 10

Element not found:

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 3

Integer: 10

List after removing: [10]

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 2

Integer: 10

List after removing: []

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 3

List is empty:

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 2

List is empty:

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 3

Invalid choice:

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 3

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 1

Integer: 10

List after adding: [10]

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 3

Integer: 10

List after adding: [10, 10]

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 2

Integer: 10

Element not found:

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 3

Integer: 10

List after removing: [10]

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 2

Integer: 10

List after removing: []

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 3

List is empty:

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 2

List is empty:

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 3

Invalid choice:

1. Add:

2. Remove:

3. Display:

4. Quit:

Enter choice: 3

Test case 2 passed

Explorer

Average time

0.072 s

71.67 ms

Maximum time

0.089 s

89.00 ms

2 out of 2 shown test case(s) passed

1 out of 1 hidden test case(s) passed

Debugger

|                           |                           |
|---------------------------|---------------------------|
| Integer: 11               | Integer: 11               |
| List after removing: [25] | List after removing: [25] |
| 1. Add                    | 1. Add                    |
| 2. Remove                 | 2. Remove                 |
| 3. Display                | 3. Display                |
| 4. Quit                   | 4. Quit                   |
| Enter choice: 2           | Enter choice: 2           |
| Integer: 25               | Integer: 25               |
| List after removing: []   | List after removing: []   |
| 1. Add                    | 1. Add                    |
| 2. Remove                 | 2. Remove                 |
| 3. Display                | 3. Display                |
| 4. Quit                   | 4. Quit                   |
| Enter choice: 3           | Enter choice: 3           |
| List is empty             | List is empty             |
| 1. Add                    | 1. Add                    |
| 2. Remove                 | 2. Remove                 |
| 3. Display                | 3. Display                |
| 4. Quit                   | 4. Quit                   |
| Enter choice: 2           | Enter choice: 2           |
| List is empty             | List is empty             |
| 1. Add                    | 1. Add                    |
| 2. Remove                 | 2. Remove                 |
| 3. Display                | 3. Display                |
| 4. Quit                   | 4. Quit                   |
| Enter choice: 8           | Enter choice: 8           |
| Invalid choice            | Invalid choice            |
| 1. Add                    | 1. Add                    |
| 2. Remove                 | 2. Remove                 |

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Reset

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|                 |                 |
|-----------------|-----------------|
| List is empty   | List is empty   |
| 1. Add          | 1. Add          |
| 2. Remove       | 2. Remove       |
| 3. Display      | 3. Display      |
| 4. Quit         | 4. Quit         |
| Enter choice: 8 | Enter choice: 8 |
| Invalid choice  | Invalid choice  |
| 1. Add          | 1. Add          |
| 2. Remove       | 2. Remove       |
| 3. Display      | 3. Display      |
| 4. Quit         | 4. Quit         |
| Enter choice: 4 | Enter choice: 4 |

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## 2.1.2. Dictionary Operations

07:23

Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

**Operations to be Performed:**

1. Insertion – Insert a new key-value pair into the dictionary using user input.
2. Update – Update the value of an existing key using user input.
3. Deletion – Delete a specified key from the dictionary using user input.
4. Traversal – Traverse the final dictionary and display all key-value pairs.

**Input and Output Format:**

1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.
2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
4. Finally, traverse the dictionary and display all key-value pairs.

The program should also display the original dictionary before performing any operations. Each output should be printed with appropriate messages.

**Note:**

- All operations must be performed using dictionary methods.
- Perform deletion only if possible; leave the dictionary unchanged.
- Refer to the visible test cases for better understanding and strictly match with the input/outputs.

# Initial dictionary with 10 predefined records

```
student = {  
    1: "Amit",  
    2: "Riya",
```

```
3: "Kiran",  
4: "Neha",  
5: "Arjun",  
6: "Pooja",  
7: "Rahul",  
8: "Sneha",  
9: "Vikram",  
10: "Anjali"  
}
```

```
# Display original dictionary  
print("Original Dictionary:", student)
```

```
# Insertion  
key = int(input())  
value = input()  
student[key] = value  
print("After Insertion:", student)
```

```
# Update  
upd_key = int(input())  
upd_value = input()  
if upd_key in student:  
    student[upd_key] = upd_value  
print("After Update:", student)
```

```
# Deletion  
del_key = int(input())  
if del_key in student:
```

```

        student.pop(del_key)

    print("After Deletion:", student)

```

# Traversal

```

print("Traversing Dictionary:")

for k, v in student.items():

    print(k, ":", v)

```

Average time: 0.027 s (27.00 ms) | Maximum time: 0.044 s (44.00 ms) | 2 out of 2 shown test case(s) passed | 2 out of 2 hidden test case(s) passed

| Expected output                                                                                                                                         | Actual output                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Original-Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}           | Original-Dictionary: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali'}           |
| 11: Suresh                                                                                                                                              | 11: Suresh                                                                                                                                              |
| After-Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'} | After-Insertion: {1: 'Amit', 2: 'Riya', 3: 'Kiran', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'} |
| 3: Karthik                                                                                                                                              | 3: Karthik                                                                                                                                              |
| After-Update: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}  | After-Update: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 5: 'Arjun', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}  |
| 5: Suresh                                                                                                                                               | 5: Suresh                                                                                                                                               |
| After-Deletion: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}            | After-Deletion: {1: 'Amit', 2: 'Riya', 3: 'Karthik', 4: 'Neha', 6: 'Pooja', 7: 'Rahul', 8: 'Sneha', 9: 'Vikram', 10: 'Anjali', 11: 'Suresh'}            |
| Traversing-Dictionary:                                                                                                                                  | Traversing-Dictionary:                                                                                                                                  |
| 1: Amit                                                                                                                                                 | 1: Amit                                                                                                                                                 |
| 2: Riya                                                                                                                                                 | 2: Riya                                                                                                                                                 |
| 3: Karthik                                                                                                                                              | 3: Karthik                                                                                                                                              |
| 4: Neha                                                                                                                                                 | 4: Neha                                                                                                                                                 |
| 6: Pooja                                                                                                                                                | 6: Pooja                                                                                                                                                |
| 7: Rahul                                                                                                                                                | 7: Rahul                                                                                                                                                |
| 8: Sneha                                                                                                                                                | 8: Sneha                                                                                                                                                |
| 9: Vikram                                                                                                                                               | 9: Vikram                                                                                                                                               |
| 10: Anjali                                                                                                                                              | 10: Anjali                                                                                                                                              |
| 11: Suresh                                                                                                                                              | 11: Suresh                                                                                                                                              |

Terminal | Test cases

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## 2.2.1. Linear search Technique

04:45

Write a program to check whether the given element is present or not in the array of elements using linear search.

**Input format:**



- The first line of input contains the array of integers which are separated by space
- The last line of input contains the key element to be searched

**Output format:**

- If the element is found, print the index.
- If the element is not found, print **Not found**.

**Sample Test Case:**

**Input:**

1 2 3 4 3 5 6

3

**Output:**

2

```
# Read input array
```

```
arr = list(map(int, input().split()))
```

```
# Read key element
```

```
key = int(input())
```

```
# Linear search
```

```
found = False
```

```

for i in range(len(arr)):
    if arr[i] == key:
        print(i)
        found = True
        break

```

# If not found

if not found:

```

    print("Not found")

```

The screenshot displays a test results interface for a coding problem. At the top, it shows performance metrics: Average time (0.010 s, 9.75 ms) and Maximum time (0.012 s, 12.00 ms). Below this, a green banner indicates '2 out of 2 shown test case(s) passed' and '2 out of 2 hidden test case(s) passed'. The main section lists two test cases, both of which passed. Test case 1 took 10 ms and Test case 2 took 12 ms. For each test case, the 'Expected output' and 'Actual output' are shown, both being '1 2 3 4 3 5 6' followed by '3' on the next line and '2' on the third line. A 'Debug' button and a menu icon are visible next to the test case headers.

## 2.2.2. Captain of the Team

01:32

You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

### Input Format:

The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

### Output Format

The output should be the height (in centimeters) of the tallest player.

```
# Read heights of 11 players
```

```
heights = list(map(int, input().split()))
```

```
# Find the tallest player
```

```
tallest = max(heights)
```

```
# Output the tallest height
```

```
print(tallest)
```

Average time  
**0.005 s**  
5.33 ms

Maximum time  
**0.006 s**  
6.00 ms

✓ 1 out of 1 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 6 ms Debug

Expected output  
171 169 185 156 174 191 186 190 187 172 160  
191

Actual output  
171 169 185 156 174 191 186 190 187 172 160  
191

Terminal

Test cases

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